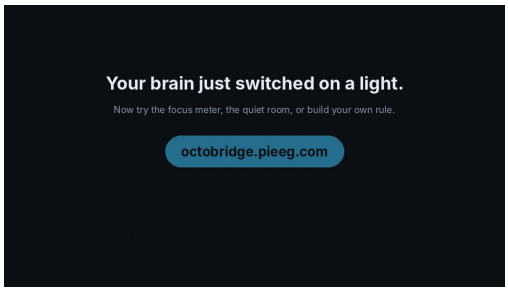
QUICK START & SAFETY GUIDE · SKU PIA24028P

From the Box to "Your Brain Switched On a Light"

Six steps in one browser tab: open the lab, flash both boards once, connect them, put on the Octopus 16, check the signal and build your first block program. Plus safety, classroom tips and troubleshooting.



WARNING Battery only on the head. Never mains on the relay.

The Octopus 16 has no isolation from its power supply. While it is on a head it runs from a small 5 V power bank only — never plugged into a computer or a wall charger. USB is for flashing only, with the board off the head. Never connect the relay to mains voltage. The OctoBio Kit is not a medical device.

01 SAFETY RULES — READ BEFORE ANYONE PUTS THE KIT ON

#	Rule
1	**Power bank only while worn.** Never wear the Octopus 16 while it is plugged into a computer or a wall charger.
2	**Flash off the head.** USB is only for flashing, with the board off the head.
3	**Relay: low voltage only.** Never connect it to mains.
4	**No diagnosis.** Focus and relax are uncalibrated learning indicators, never a measure of mood or ability.
5	**Implanted devices.** Ask a physician first if someone has a pacemaker or other implant.
6	**Skin and hygiene.** Stop if the skin becomes irritated; clean the pogo pins with isopropyl alcohol between students.

Power source for the Octopus 16	While worn	Why
Small 5 V power bank	SAFE	Own battery, no path to the mains
USB cable to a computer	NEVER	For flashing only
Wall charger	NEVER	Connects the person to the mains

02 WHAT YOU NEED

Item	Requirement
OctoBio Kit	Octopus 16 with XIAO ESP32-S3, starter board in its case
Computer	Bluetooth, desktop Chrome or Edge (not Safari, Firefox or phones)
5 V power bank	Small, for the Octopus 16
USB data cables	USB-C, for flashing and for the starter board
Strap or headband	To hold the Octopus 16 on the back of the head

03 SETUP IN SIX STEPS

1 Open the lab

Go to **octobridge.pieeg.com** in desktop Chrome or Edge. No account, no install.

2 Flash both boards once

Open **Flash**. Octopus 16 (off the head): plug in the XIAO, click **Install Octopus-16**; no port? hold BOOT, tap RESET, release BOOT. Starter board: close other serial monitors, click **Flash octoBio Kit** (~20 s) — the LCD shows "octoBio Kit Ready".

3 Connect

Open **Connect**. Power the Octopus 16 from the power bank and pick **Octopus-16-XXXX**. Plug in the starter board and pick its port; it restarts — wait for **live**.

4 Put it on

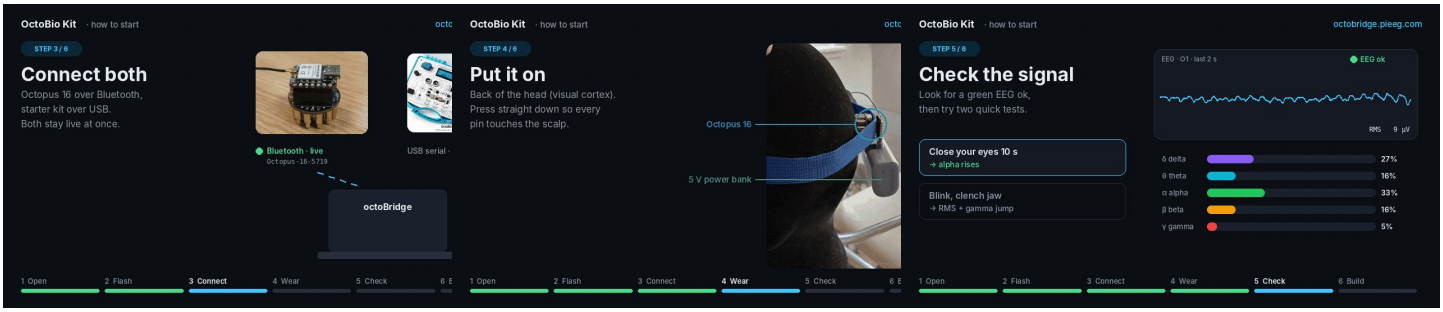
Strap the Octopus 16 over the **back of the head**, part the hair and press it straight down so every pin touches the scalp.

5 Check the signal

On **Board**, "EEG ok" turns green (RMS 2–200 µV). Close the eyes 10 s: alpha rises. Blink and clench the jaw: RMS and gamma jump — that is muscle, not a thought.

6 Build

Open **Blocks**, stack **forever → if alpha is high → light D10 on / else off** and press **Run**. Close your eyes and the LED turns on.



Connect both devices. Put it on. Check the signal.

> **TIP No headset yet?**
> Start with Lesson 14, **Follow the knob**: knob → servo, starter board only. Teach Lesson 24, **Blink Is Not a Thought**, early.

04 TROUBLESHOOTING

Connecting and flashing

Problem	Fix
The lab will not connect	Desktop Chrome or Edge on the HTTPS site; Safari, Firefox and phones are not supported.
No serial port	Use a USB data cable; close the Arduino IDE and other serial monitors.
XIAO port not found	Hold BOOT, tap RESET, release BOOT, retry.
Octopus silent after flashing	Unplug and replug the USB once.
Board numbers do not move	The board restarts when the port opens — wait for "live".

Signal and readings

Problem	Fix
Octopus on Bluetooth but no EEG, LED double-blinks	XIAO is seated the wrong way round — reseat with 5V / GND / 3V3 aligned.
"EEG ok" stays red	Press straight down so every pin touches the scalp; sit still; clean the pins with IPA.
Connects then drops	Retry, move closer, sit still for the first run.
Distance is negative	The echo was missed — point the sensor at a facing surface.
A pogo pin is stuck	A drop of IPA in the barrel, gently work the plunger, let it dry. No oil or grease.

05 RELATED DOCUMENTS

OB-DS-001 Product Datasheet	**OB-MD-002** Mechanical & Layout Drawing	**OB-PCB-003** Hardware & Manufacturing Specification
OB-PIN-004 Pinout & Connection Guide	**OB-SW-006** Software & Protocol Guide	